DJANGO PROJECT SETUP ON UBUNTU 22.4(CLEAN & CORRECT)

Step 1: Create EC2 Instance

- **Ubuntu 22.04 LTS**
- **Storage: 30 GB** (fine)
- Open **ports 22, 8000 (temporary), 80** in Security Group

- If port 80 isn't open later, Nginx won't work. Don't forget this.

Step 2: Update System

sudo apt update && sudo apt upgrade -y

```
ubuntu@ip-172-31-32-181:~$ sudo apt update && sudo apt upgrade -y
Hit:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:4 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:5 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3188 kB]
Get:6 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1250 kB]
Fetched 4821 kB in 1s (4339 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
64 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following NEW packages will be installed:
  linux-aws-6.8-headers-6.8.0-1044 linux-aws-6.8-tools-6.8.0-1044 linux-headers-6.8.0-1044-aws linux-image-6.8.0-1044-aws linux-modules-6.8.0-1044-aws
  linux-tools-6.8.0-1044-aws
The following packages have been kept back:
  ubuntu-minimal ubuntu-server ubuntu-standard
The following packages will be upgraded:
  apparmor bind9-dnsutils bind9-host bind9-libs binutils binutils-common binutils-x86-64-linux-gnu dirmngr distro-info-data gnupg gnupg-l10n gnupg-utils
```

Step 3: Install Required Packages

sudo apt install python3-pip python3-venv nginx -y

```
root@ip-172-31-29-149:/home/ubuntu# sudo apt install python3-pip python3-venv nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  build-essential bzip2 cpp cpp-11 dpkg-dev fakeroot fontconfig-config fonts-dejavu-core g++ g++-11 gcc gcc-11 gcc-11-base javascript-common libalgorithm-diff-perl
  libalgorithm-diff-xs-perl libalgorithm-merge-perl libasan6 libatomic1 libc-dev-bin libc-devtools libc6-dev libcc1-0 libcrypt-dev libdeflate0 libdpkg-perl libexpat1-dev libfakeroot
  libfile-fcntllock-perl libfontconfig1 libgcc-11-dev libgd3 libgomp1 libisl23 libitm1 libjbig0 libjpeg-turbo8 libjpeg8 libjs-jquery libjs-sphinxdoc libjs-underscore liblsan0 libmpc3
  libnginx-mod-http-geoip2 libnginx-mod-http-image-filter libnginx-mod-http-xslt-filter libnginx-mod-mail libnginx-mod-stream libnginx-mod-stream-geoip2 libnsl-dev libpython3-dev
  libpython3.10-dev libquadmath0 libstdc++-11-dev libtiff5 libtirpc-dev libtsan0 libubsan1 libwebp7 libxpm4 linux-libc-dev lto-disabled-list make manpages-dev nginx-common nginx-core
  python3-dev python3-pip-whl python3-setuptools-whl python3-wheel python3.10-dev python3.10-venv rpcsvc-proto zlib1g-dev
Suggested packages:
  bzip2-doc cpp-doc gcc-11-locales debian-keyring g++-multilib g++-11-multilib gcc-11-doc gcc-multilib autoconf automake libtool flex bison gdb gcc-doc gcc-11-multilib glibc-doc bzr
  libgd-tools libstdc++-11-doc make-doc fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  build-essential bzip2 cpp cpp-11 dpkg-dev fakeroot fontconfig-config fonts-dejavu-core g++ g++-11 gcc gcc-11 gcc-11-base javascript-common libalgorithm-diff-perl
  libalgorithm-diff-xs-perl libalgorithm-merge-perl libasan6 libatomic1 libc-dev-bin libc-devtools libc6-dev libcc1-0 libcrypt-dev libdeflate0 libdpkg-perl libexpat1-dev libfakeroot
  libfile-fcntllock-perl libfontconfig1 libgcc-11-dev libgd3 libgomp1 libisl23 libitm1 libjbig0 libjpeg-turbo8 libjpeg8 libjs-jquery libjs-sphinxdoc libjs-underscore liblsan0 libmpc3
  libnginx-mod-http-geoip2 libnginx-mod-http-image-filter libnginx-mod-http-xslt-filter libnginx-mod-mail libnginx-mod-stream libnginx-mod-stream-geoip2 libnsl-dev libpython3-dev
  libpython3.10-dev libquadmath0 libstdc++-11-dev libtiff5 libtirpc-dev libtsan0 libubsan1 libwebp7 libxpm4 linux-libc-dev lto-disabled-list make manpages-dev nginx nginx-common
  nginx-core python3-dev python3-pip python3-pip-whl python3-setuptools-whl python3-venv python3-wheel python3.10-dev python3.10-venv rpcsvc-proto zlib1g-dev
0 upgraded, 77 newly installed, 0 to remove and 3 not upgraded.
Need to get 74.5 MB of archives.
```

i-01468d57d246b5b91 (Django)

PublicIPs: 100.48.89.121 PrivateIPs: 172.31.29.149

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Step 4: Check Python Version

python3 –version

```
ubuntu@ip-172-31-32-181:~$ python3 --version
Python 3.10.12
```

Step 5: Create Project Directory

mkdir myproject

cd myproject

```
ubuntu@ip-172-31-32-181:~$ mkdir myproject
ubuntu@ip-172-31-32-181:~$ cd myproject
```

Step 6: Create Virtual Environment

python3 -m venv venv

source venv/bin/activate

```
ubuntu@ip-172-31-32-181:~/myproject$ python3 -m venv venv
ubuntu@ip-172-31-32-181:~/myproject$ source venv/bin/activate
(venv) ubuntu@ip-172-31-32-181:~/myproject$
```

Step 7: Install Django & Gunicorn

pip install django gunicorn

```
(venv) ubuntu@ip-172-31-32-181:~/myproject$ pip install django gunicorn
Collecting django
  Downloading django-5.2.10-py3-none-any.whl (8.3 MB)
     ---------------------------------------- 8.3/8.3 MB 35.7 MB/s eta 0:00:00
Collecting gunicorn
  Downloading gunicorn-24.1.1-py3-none-any.whl (114 kB)
     ---------------------------------------- 114.9/114.9 KB 19.4 MB/s eta 0:00:00
Collecting asgiref>=3.8.1
  Downloading asgiref-3.11.0-py3-none-any.whl (24 kB)
Collecting sqlparse>=0.3.1
  Downloading sqlparse-0.5.5-py3-none-any.whl (46 kB)
     ---------------------------------------- 46.1/46.1 KB 6.2 MB/s eta 0:00:00
Collecting packaging
  Downloading packaging-26.0-py3-none-any.whl (74 kB)
     ---------------------------------------- 74.4/74.4 KB 12.3 MB/s eta 0:00:00
Collecting typing_extensions>=4
  Downloading typing_extensions-4.15.0-py3-none-any.whl (44 kB)
     ---------------------------------------- 44.6/44.6 KB 6.6 MB/s eta 0:00:00
Installing collected packages: typing_extensions, sqlparse, packaging, gunicorn, asgiref, django
Successfully installed asgiref-3.11.0 django-5.2.10 gunicorn-24.1.1 packaging-26.0 sqlparse-0.5.5 typing_extensions-4.15.0
(venv) ubuntu@ip-172-31-32-181:~/myproject$
```

Step 8: Create Django Project

django-admin startproject mysite

cd mysite

```
(venv) ubuntu@ip-172-31-32-181:~/myproject$ django-admin startproject mysite
(venv) ubuntu@ip-172-31-32-181:~/myproject$ cd mysite
```

Step 9: Verify Structure

ls

Expected:

manage.py mysite


```
(venv) ubuntu@ip-172-31-32-181:~/myproject$ django-admin startproject mysite
(venv) ubuntu@ip-172-31-32-181:~/myproject$ cd mysite
```


```
(venv) ubuntu@ip-172-31-32-181:~/myproject/mysite$ ls
manage.py  mysite
(venv) ubuntu@ip-172-31-32-181:~/myproject/mysite$
```

Step 10: Inspect Inner Project Folder

cd mysite

ls

Expected:

__init__.py asgi.py settings.py urls.py wsgi.py

```
(venv) ubuntu@ip-172-31-32-181:~/myproject/mysite/mysite$ ls
__init__.py  asgi.py  settings.py  urls.py  wsgi.py
```

Step 11: Database Migrations

python manage.py migrate

```
(venv) ubuntu@ip-172-31-32-181:~/myproject/mysite$ python manage.py migrate
Operations to perform:
  Apply all migrations: admin, auth, contenttypes, sessions
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying sessions.0001_initial... OK
(venv) ubuntu@ip-172-31-32-181:~/myproject/mysite$
```

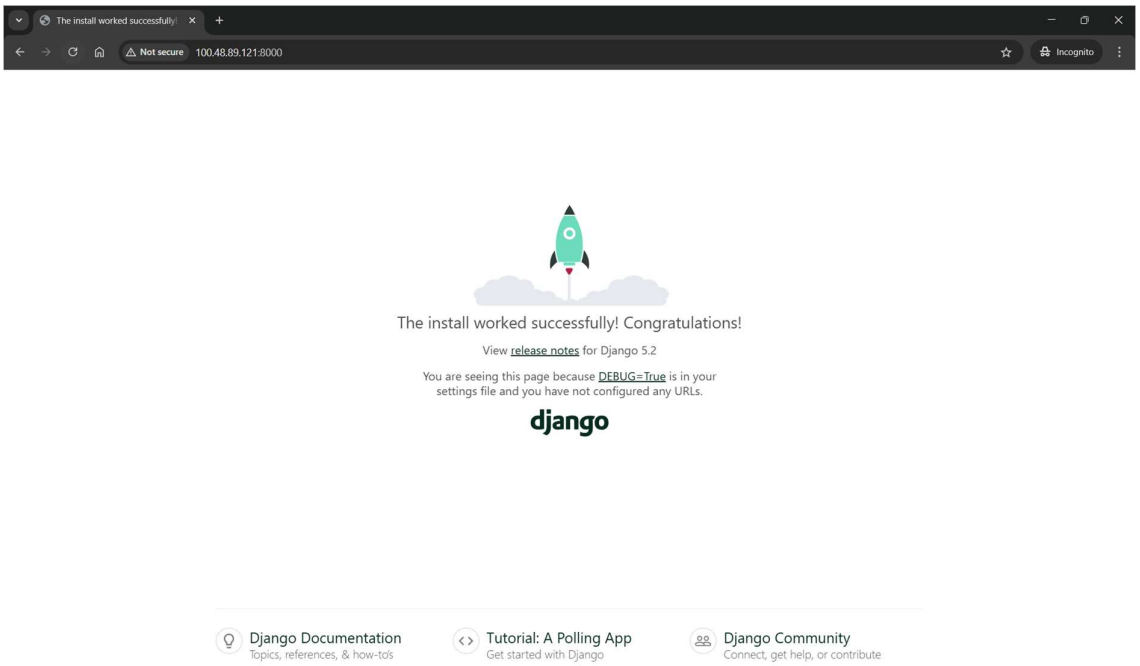
Step 12: Run Development Server

python manage.py runserver 0.0.0.0:8000

http://<EC2_PUBLIC_IP>:8080

```
(venv) ubuntu@ip-172-31-32-181:~/myproject/mysite$ python manage.py runserver 0.0.0.0:8000
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).
February 01, 2026 - 06:04:23
Django version 5.2.10, using settings 'mysite.settings'
Starting development server at http://0.0.0.0:8000/
Quit the server with CONTROL-C.

WARNING: This is a development server. Do not use it in a production setting. Use a production WSGI or ASGI server instead.
For more information on production servers see: https://docs.djangoproject.com/en/5.2/howto/deployment/
Invalid HTTP_HOST header: '51.21.170.69:8000'. You may need to add '51.21.170.69' to ALLOWED_HOSTS.
Traceback (most recent call last):
  File "/home/ubuntu/myproject/venv/lib/python3.10/site-packages/django/core/handlers/exception.py", line 55, in inner
```

The hosted site is…



Step 13: Configure ALLOWED_HOSTS (IMPORTANT)

Edit:

nano mysite/settings.py

ALLOWED_HOSTS = ['*'] "or"

ALLOWED_HOSTS = ['54.241.42.66', 'localhost', '127.0.0.1']

```
# Quick-start development settings - unsuitable for production
# See https://docs.djangoproject.com/en/5.2/howto/deployment/checklist/

# SECURITY WARNING: keep the secret key used in production secret!
SECRET_KEY = 'django-insecure-hl7h0yn^*+t^-#ewd_0&21rc+anlx=o$fjdb5jqv1pm%!dr05p'

# SECURITY WARNING: don't run with debug turned on in production!
DEBUG = True

ALLOWED_HOSTS = ['*']


# Application definition

INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
```

Step 14: Restart Server

python manage.py runserver 0.0.0.0:8000

Step 16: Create Superuser

python manage.py createsuperuser

Enter:

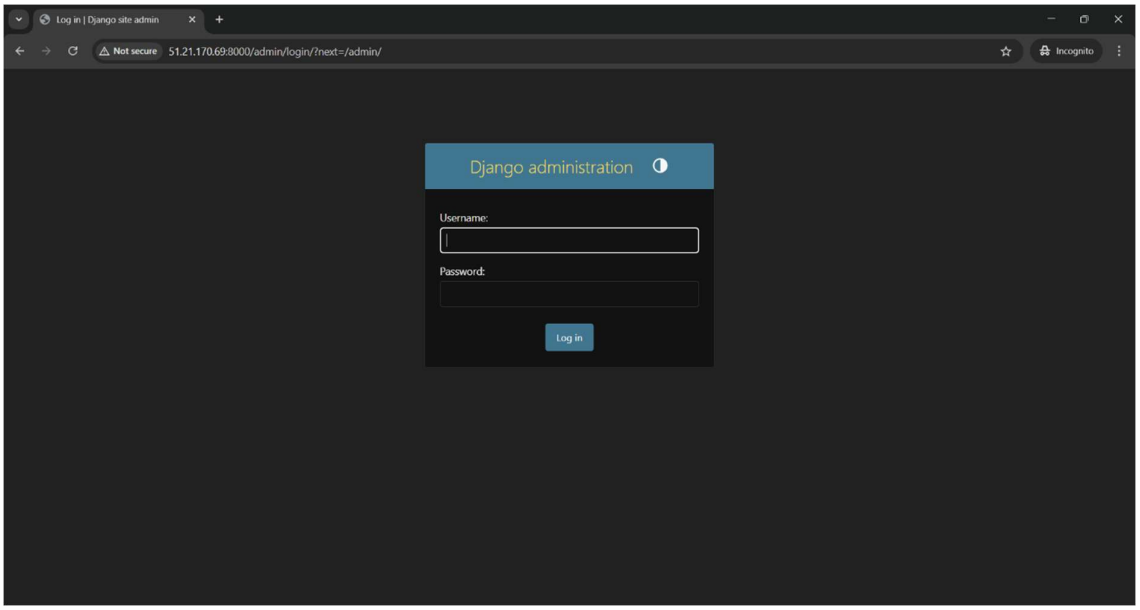
- **Username**
- **Email**
- **Password**

Login:

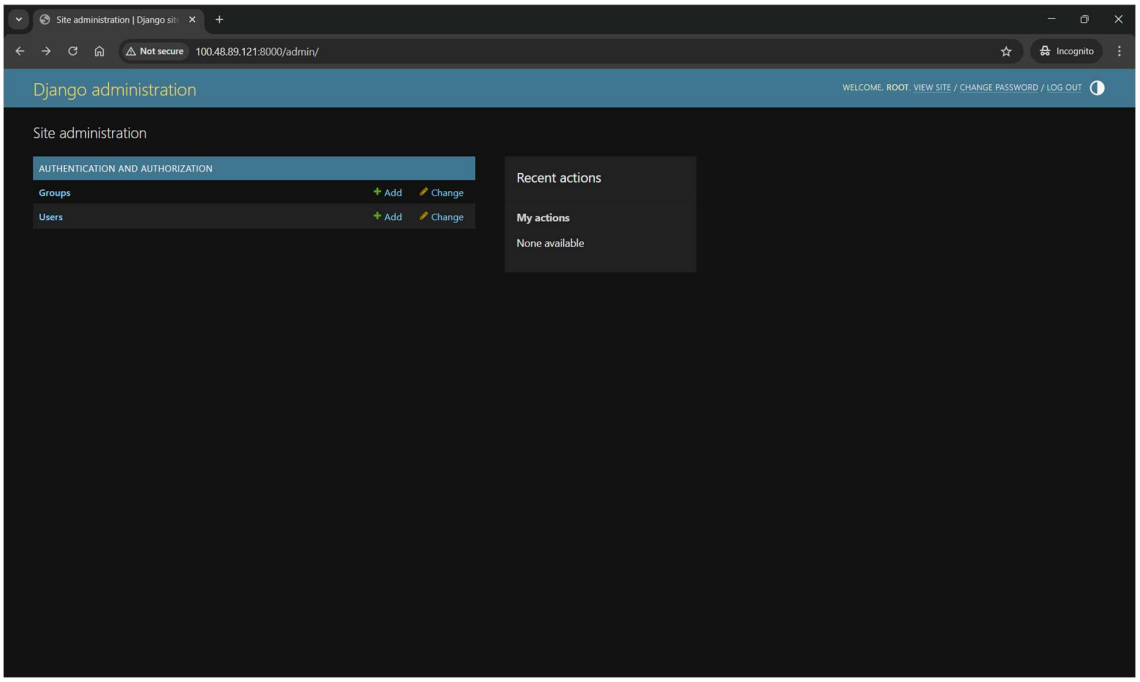
http://<EC2_PUBLIC_IP>:8080/admin

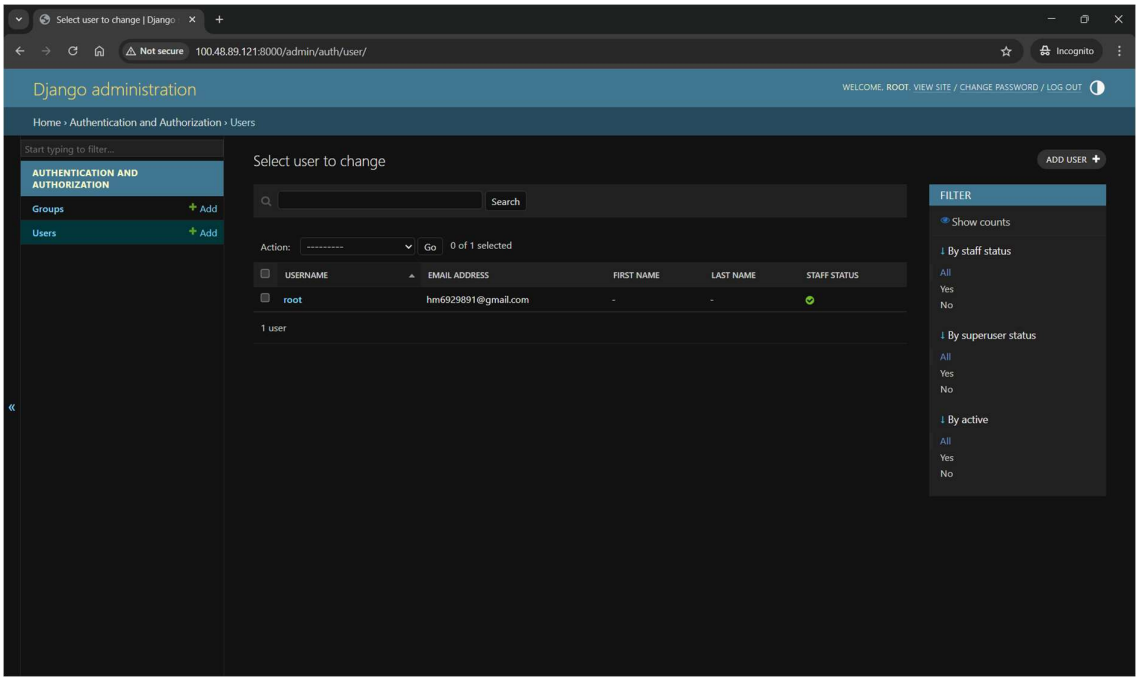
```
[31/Jan/2026 06:47:17] "GET / HTTP/1.1" 200 12068
Not Found: /favicon.ico
[31/Jan/2026 06:47:17] "GET /favicon.ico HTTP/1.1" 404 2212
^C(venv) root@ip-172-31-29-149:/home/ubuntu/myproject/mysite# python manage.py createsuperuser
Username (leave blank to use 'root'):
Email address: hm6929891@gmail.com
Password:
Password (again):
This password is too short. It must contain at least 8 characters.
Bypass password validation and create user anyway? [y/N]: y
Superuser created successfully.
(venv) root@ip-172-31-29-149:/home/ubuntu/myproject/mysite# python manage.py runserver 0.0.0.0:8000
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).
January 31, 2026 - 06:50:10
Django version 5.2.10, using settings 'mysite.settings'
Starting development server at http://0.0.0.0:8000/
Quit the server with CONTROL-C.
```

You have to login with the superuser we have created.



The admin page…




Select user to change | Django
Not secure
100.48.89.121:8000/admin/auth/user/
Incognito
Django administration
WELCOME, ROOT. VIEW SITE / CHANGE PASSWORD / LOG OUT
Home › Authentication and Authorization › Users
Start typing to filter...
AUTHENTICATION AND AUTHORIZATION
Groups
Add
Users
Add
Select user to change
ADD USER
Search
Action:

Go
0 of 1 selected
USERNAME
EMAIL ADDRESS
FIRST NAME
LAST NAME
STAFF STATUS
root
hm6929891@gmail.com
-
-
1 user
FILTER
Show counts
By staff status
All
Yes
No
By superuser status
All
Yes
No
By active
All
Yes
No